

BUKUNMI ENOCH AKINYEMI

AI Engineer · Agents, Distributed Systems & Applied ML

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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFILE

AI engineer who ships production systems end to end - event-driven backends, tool-using agents, and applied ML. Recent independent builds: Tideo, a distributed video-transcoding pipeline (Kafka, Celery, Redis, PostgreSQL), and Aristotle, a source-aware research agent with web search (SearXNG) and sandboxed code execution. Nearly two years of AI engineering in legal technology - a domain where every citation is checkable and hallucination is a liability - most recently at Qanooni AI: document-processing pipelines, a Word-embedded agent, and English/Arabic evaluation. First Class law degree behind the domain depth.

EXPERIENCE

Qanooni AI · AI Engineer - Dubai-based, Remote

October 2025 – June 2026

Built and shipped production LLM systems - agents, document pipelines, and evaluation harnesses - in a regulated, correctness-critical domain.

- Built QCounsel, a multi-tool Pydantic AI agent running inside Microsoft Word for legal research, drafting, and litigation workflows; designed its evaluation framework, tool-routing classifier, and clarify-vs-act logic.
- Shipped multilingual evaluation infrastructure: English and Arabic evaluation splits, jurisdiction-aware LLM-as-judge prompts, pipeline evaluations, and prompt optimization.
- Integrated DocETL pipeline orchestration for document processing at scale (Split → ParallelMap → CodeMap → Reduce).
- Production AI ops: triaged bugs from Langfuse trace IDs; span-level latency debugging; PostHog vs database telemetry reconciliation for apply-rate analysis.
- Stack: Python, FastAPI, Pydantic AI, DocETL, Langfuse, Anthropic & OpenAI API, pytest, Ruff, Pyright.

CaseSimpli Legal Tech Solutions · Lead Software Engineer - Remote

November 2024 – October 2025

Led engineering on an AI-assisted document automation platform as sole engineer.

- Built the document automation system end to end, cutting contract drafting time by 30% while preserving human-in-the-loop review.
- Built retrieval-augmented generation pipelines over statutory and precedent corpora, with citation grounding and controlled, schema-constrained outputs.
- Developed a domain-specific conversational assistant; improved case-analysis efficiency by 25%.
- Worked directly with domain experts to align system behavior with substantive correctness requirements.
- Stack: Python (FastAPI), Next.js, PostgreSQL, OpenAI Assistants, Azure.

Data Quotient Hub · Frontend Web Engineer (Part-Time)

April 2025 – October 2025

Part-time frontend engineering alongside full-time backend and AI work.

- Shipped responsive enterprise UIs in Next.js + TypeScript for a data-management product; integrated backend APIs and improved client-side performance.

Glory Schools, Egbedi · ICT Manager & Instructor

2022 – 2024

- Designed, built, and operated the school's full management platform - examinations, e-learning, digital library, messaging, payments, admissions, and role-gated administration - with web and mobile clients (99.9% uptime).
 - Stack: Python (FastAPI) + React/TypeScript + PostgreSQL + Docker; JWT authentication and role-based access control; 1,400+ books and 3,000+ videos served.
 - Taught practical programming and ICT alongside running the platform.
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SELECTED PROJECTS

Tideo - Distributed video transcoding pipeline · [Live demo](#) · [GitHub](#) - Turns one source file into a full HLS ladder: every rendition encoded in parallel across Celery workers, with Kafka event facts, RabbitMQ command dispatch, Redis hot state, and PostgreSQL as durable audit state. Built for failure - idempotent event envelopes, dead-letter queues, deduplication, backpressure, and retry-after handling, with 48 test modules covering them. Ships poster frames, scrubbable storyboards, an embed player, and optional captions.

Aristotle - Source-aware AI research assistant · [Live demo](#) · [GitHub](#) - Four independently deployable services (agent API, model gateway, search, isolated Python sandbox) behind a React client with WebSocket streaming. Tool-using Pydantic AI agent with model fallback, cancellable runs, and sandboxed code execution over uploaded documents; ships with its own eval suite and three CI pipelines that deploy the services separately.

Capit - Glass-box image captioner · [Live demo](#) · [GitHub](#) · [Model \(HF\)](#) - Show, Attend and Tell trained from scratch (frozen ResNet-50 encoder, Bahdanau attention + LSTM decoder, Flickr8k): BLEU-4 23.63 / CIDEr 62.80 (beam 5, Karpathy split). UI exposes per-word attention heatmaps and the beam candidates it rejected, with BLIP alongside as a closed-box contrast. Deployed on an HF Space serving a self-contained model artifact; 21-module test suite over the training pipeline, including single-batch overfit gates.

Givemore - MovieLens recommender · [Live demo](#) · [GitHub](#) - Item-item collaborative filtering (adjusted cosine, IUF-weighted) blended with TF-IDF content similarity and a Bayesian popularity fallback - precomputed offline into a 9 MB SQLite artifact served by a zero-ML-dependency FastAPI; framework-free TypeScript frontend.

LLM Citation-Hallucination Benchmark · [GitHub](#) - Async LLM evaluation pipeline measuring citation hallucination across 6 frontier models × 328 prompts (English + Arabic). Pluggable providers, pydantic schemas, citation-fabrication rubric, resume capability. Modeled on Stanford RegLab's legal_hallucinations.

ML from First Principles · [GitHub](#) - micrograd and makemore rebuilt from scratch (reverse-mode autograd, tokenization, transformer language models), maintained continuously alongside MSc financial-econometrics research.

EDUCATION

ESCAE Benin University · LLB Law, First Class Honours (2018–2022) - GPA 4.92/5.0, top 1% of cohort; Best Graduating Student, Faculty of Law.

WorldQuant University · MSc Financial Engineering (2025–present) - financial econometrics: cointegration, ECM/VECM, LASSO, Diebold-Mariano testing.

University of the People · BSc Computer Science (2026–present)

TECHNICAL

AI & ML: Agentic systems & tool use · RAG pipelines · LLM evaluation and LLM-as-judge · Model training from scratch (PyTorch) · Recommender systems · Langfuse · DocETL · LangChain · Hugging Face · Claude API · OpenAI API

Languages & Frameworks: Python (FastAPI, SQLAlchemy, Pydantic, Celery, pytest) · TypeScript (React, Next.js, Vite) · SQL

Distributed & Event-Driven: Kafka · RabbitMQ · Celery · Redis · event-driven architecture · idempotency, dead-letter queues, backpressure, rate limiting

Data & Infrastructure: PostgreSQL · SQLite · Chroma · FAISS · Docker · CI/CD (GitHub Actions) · Caddy · Azure · AWS

CERTIFICATIONS

Fundamentals of LLMs (DeepLearning.AI) · Advanced Learning Algorithms and Unsupervised Learning, Recommenders & RL (Coursera ML Specialization) · Oracle Cloud Infrastructure 2025 AI Foundations Associate · Applied Data Science Lab and Foundations of Financial Engineering (WorldQuant University)

LANGUAGES

English (Full Professional Proficiency) · Yoruba (Native)